Compiled Solutions – Senior Division

1. **00-01 C3 Digital Electronics**

$\overline{(\overline{A}B+\overline{B})\oplus\overline{B+\overline{C}}}$

(0,0,1) (1,0,1) (1,1,0) (1,1,1)

A	B	C	$\overline{A}$	$\overline{B}$	$\overline{C}$	$\overline{A}B$	$\overline{A}B+\overline{B}$	$B+\overline{C}$	$\overline{B+\overline{C}}$	$\overline{\oplus}$ (both the same)
0	0	0	1	1	1	0	1	1	0	0
0	0	1	1	1	0	0	1	0	1	1
0	1	0	1	0	1	1	1	1	0	0
0	1	1	1	0	0	1	1	1	0	0
1	0	0	0	1	1	0	1	1	0	0
1	0	1	0	1	0	0	1	0	1	1
1	1	0	0	0	1	0	0	1	0	1
1	1	1	0	0	0	0	0	1	0	1

2. **00-01 C3 Digital Electronics**

The translation is as follows: $\overline{(A\oplus B)(\overline{B\oplus C})}$

3. **00-01 C4 Digital Electronics**

The circuit translates as follows: $(A+AB+(\overline{B}+C))(\overline{B}+C)(\overline{C+D})$ 2

To be true all 3 factors must be TRUE.

$\overline{C+D}$ is only TRUE for the quadruple (*, *, 0, 0),

but when B = 1, the second factor is FALSE.

Therefore, the choices are limited to (*, 0, 0, 0).

4. **01-02 C3 Digital Electronics**

The circuit translates as follows: $\overline{A\overline{B}+C}$. This simplifies to: (*,1,0) and (0,0,0)

$\overline{A}\overline{\overline{B}}\overline{C}=(\overline{A}+B)\overline{C}$. To be TRUE, both factors must be TRUE. OR

This implies that C must equal 0. The first factor gives 2 choices. (0,1,0), (1,1,0),

If A = 0, then B can be either 1 or 0. If A = 1, then B must equal 1. and (0, 0, 0).

5. **01-02 C3 Digital Electronics**

The circuit translates to: $(\overline{A+B})((B\oplus C)D)$. If B is FALSE, $\overline{A}CD$

$(\overline{A+B})=\overline{A}$, $B\oplus C=C$ so the entire expression is $\overline{A}CD$.

6. **01-02 C4 Digital Electronics**

The circuit translates: $\overline{(AB+\overline{C})(\overline{AB})}$ (0,0,1), (0,1,1),
To be true one or both of the terms must be true. This occurs for (1,0,1), (1,1,1),
(*, *, 1) and (1, 1, 0). (1,1,0)

7. **02-03 C3 Digital Electronics**

The circuit translates as follows: $\overline{(AB\oplus(B+\overline{C}))C}$ (0, 1, 0)

$\overline{(AB\oplus(B+\overline{C}))C}=0$ implies $(AB\oplus(B+\overline{C}))C=1$ $\overline{C}=1$ so $C=0$. The XOR expression must = 1 also. If B = 0 both operands are FALSE and the expression is FALSE which leads to a contradiction. If B = 1, AB must equal 0. Since B = 1, A must = 0.

8. **02-03 C3 Digital Electronics**

The circuit translates as follows: $\overline{A+(\overline{AB}\oplus C)}$ 6

A	B	C	AB	$\overline{AB}$	$\oplus$ C	+ A	NOT
0	0	0	0	1	1	1	0
0	0	1	0	1	0	0	1
0	1	0	0	1	1	1	0
0	1	1	0	1	0	0	1
1	0	0	0	1	1	1	0
1	0	1	0	1	0	1	0
1	1	0	1	0	0	1	0
1	1	1	1	0	1	1	0

There are 6 ordered triples that make it FALSE.

9. **02-03 C4 Digital Electronics**

The three inputs going into the box are: (0,0,1), (0,1,0), (0,1,1)
$\overline{A}+AB$, $\overline{AB}$ and $B+C$

A	B	C	AB	~A	$\overline{A}+AB$	$\overline{AB}$	$B+C$
0	0	0	0	1	1	1	0
0	0	1	0	1	1	1	1
0	1	0	0	1	1	1	1
0	1	1	0	1	1	1	1
1	0	0	0	0	0	1	0
1	0	1	0	0	0	1	1
1	1	0	1	0	1	0	1
1	1	1	1	0	1	0	1

Only (0,0,1), (0,1,0), and (0,1,1) produce all 1's.

10. 03-04 C3 Digital Electronics

The circuit translates to: $\sim(\overline{A+B} \oplus \overline{BC})C$ 3

A	B	C	$A+B$	$\overline{A+B}$	$\overline{BC}$	$\oplus$	$\overline{\overline{A+B} \oplus \overline{BC}}$	$AND\ C$
0	0	0	0	1	1	0	1	0
0	0	1	0	1	1	0	1	1
0	1	0	1	0	1	1	0	0
0	1	1	1	0	0	0	1	1
1	0	0	1	0	1	1	0	0
1	0	1	1	0	1	1	0	0
1	1	0	1	0	1	1	0	0
1	1	1	1	0	0	0	1	1

11. 03-04 C3 Digital Electronics

$\overline{A+B+\overline{BC}} = 1$ so $A+B+\overline{BC} = 0$ implies that A = 0, B = 0 and $\overline{BC} = 0$. 0
This implies that $BC = 1$. That implies B = 1 and C = 1. This causes a contradiction since B must equal 0 and B must equal 1. Therefore there are no triples that make the circuit TRUE.

12. 03-04 C4 Digital Electronics

The circuit translates to $\overline{(\overline{A} \oplus \overline{(A+B)+B\overline{C}})\overline{C}}$. Only (0,1,0) makes it FALSE. (0,1,0)

A	B	C	$\overline{A}$	$A+B$	$B\overline{C}$	+	NOT	$\overline{A} \oplus$	$*\overline{C}$	NOT
0	0	1	1	0	0	0	1	0	0	1
0	0	0	1	0	0	0	1	0	0	1
0	1	1	1	1	0	1	0	1	0	1
0	1	0	1	1	1	1	0	1	1	0
1	0	1	0	1	0	1	0	0	0	1
1	0	0	0	1	0	1	0	0	0	1
1	1	1	0	1	0	1	0	0	0	1
1	1	0	0	1	1	1	0	0	0	1

13. 04-05 C3 Digital Electronics

The circuit translates to: $\overline{A}B \oplus (B+C)$. There are 4 that make it TRUE. 4

A	B	C	$\overline{A}B$	$B+C$	$\oplus$
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	0	1	1

14. 04-05 C3 Digital Electronics

The circuit translates directly to: $\overline{\overline{A} + \overline{A \oplus B} + \overline{A \oplus B}\overline{B\overline{C}} + C}$ As shown

Because of the order of operations no parentheses are needed.

An AND (*) operator may be used before the last term.

15. 04-05 C4 Digital Electronics

The circuit translates to: $(\overline{A} \oplus (\overline{A + B} * \overline{BC})) + C$. (0,0,0), (1,0,0),

Three entries make it FALSE. (1,1,0)

A	B	C	$\overline{A}$	$\overline{A+B}$	$\overline{BC}$	*	$\oplus$	$+C$
0	0	0	1	1	1	1	0	0
0	0	1	1	1	1	1	0	1
0	1	0	1	0	1	0	1	1
0	1	1	1	0	0	0	1	1
1	0	0	0	0	1	0	0	0
1	0	1	0	0	1	0	0	1
1	1	0	0	0	1	0	0	0
1	1	1	0	0	0	0	0	1

16. 05-06 C3 Digital Electronics

Sinplifying to: $\overline{\overline{A \oplus B} * (B \oplus C) + \overline{CD}}$ and using a table gives 3 TRUE results. 3

A	B	C	D	A⊕B	NOT	B⊕C	AND	NOT (CD)	+	NOT
0	0	0	0	0	1	0	0	1	1	0
0	0	0	1	0	1	0	0	1	1	0
0	0	1	0	0	1	1	1	1	1	0
0	0	1	1	0	1	1	1	0	1	0
0	1	0	0	1	0	1	0	1	1	0
0	1	0	1	1	0	1	0	1	1	0
0	1	1	0	1	0	0	0	1	1	0
0	1	1	1	1	0	0	0	0	0	1
1	0	0	0	1	0	0	0	1	1	0
1	0	0	1	1	0	0	0	1	1	0
1	0	1	0	1	0	1	0	1	1	0
1	0	1	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1	0
1	1	0	1	0	1	1	1	1	1	0
1	1	1	0	0	1	0	0	1	1	0
1	1	1	1	0	1	0	0	0	0	1

17. 05-06 C3 Digital Electronics

The circuit translates to: $\overline{\overline{A} + AB(B + C)C}$ As shown

18. 05-06 C4 Digital Electronics

The circuit translates to $((\overline{AB})(B+\overline{C+D}))+\overline{C+D}=1\Rightarrow$ 7

$\overline{A}B+\overline{A}\,\overline{C}\,\overline{D}+\overline{B}\,\overline{C}\,\overline{D}+\overline{C}\,\overline{D}=1\Rightarrow \overline{A}B+\overline{C}\,\overline{D}=1$. If A=0 and B=1 then C=* and D=*. If C=0 and D=0 then A=* and B=*. The ordered quadruplets are (0,1,*,*) and (*,*,0,0). Each produces 4 results with one overlap (0,1,0,0).

19. 06-07 C3 Digital Electronics

The circuit translates to $(\overline{A}+(A\oplus B))(\overline{B}+C)=1$. 5

Both terms of the AND must be TRUE**.**

If A=1 then B=0 and C=* $\Rightarrow$ (1,0,*)

If A=0, and B=1, then C=1, but if A=0 and B=0 then C=* $\Rightarrow$

(0,1,1) and (0,0,*). Therefore there are 5 solutions.

20. 06-07 C3 Digital Electronics

The circuit translates to $\overline{\overline{\overline{A}+AB}+(B+\overline{C})}=\overline{(\overline{A}+AB)}(\overline{B+\overline{C}})=$ $\overline{A}\,\overline{B}C$

$(\overline{A}+AB)(\overline{B}C)=\overline{A}\,\overline{B}C+AB\overline{B}C=\overline{A}\,\overline{B}C$

21. 06-07 C4 Digital Electronics

$\overline{\overline{\overline{A+\overline{\overline{A}B}}}((B+C)\oplus\overline{C})}=0$ (0,1,1)

$\overline{(A+\overline{\overline{A}B})}((B+C)\oplus\overline{C})=1$ Both terms must be TRUE

$A+\overline{\overline{A}B}=0$ AND $((B+C)\oplus\overline{C})=1$

$A=0$ AND $\overline{\overline{A}B}=0$ $\qquad (1+C)\oplus\overline{C}=1$

$\overline{A}B=1$ $\qquad 1\oplus\overline{C}=1$

$B=1$ $\qquad \overline{C}=0\Rightarrow C=1$

22. 07-08 C3 Digital Electronics

The circuit translates to $\overline{A(\overline{A}B\oplus(B+C))}+\overline{C}$. Six of them are TRUE. 6

A	B	C	$\overline{A}B$	$B+C$	$\overline{A}B\oplus(B+C)$	$\overline{A(\overline{A}B\oplus(B+C))}$	$+\overline{C}$
0	0	0	0	0	0	1	1
0	0	1	0	1	1	1	1
0	1	0	1	1	0	1	1
0	1	1	1	1	0	1	1
1	0	0	0	0	0	1	1
1	0	1	0	1	1	0	0
1	1	0	0	1	1	0	1
1	1	1	0	1	1	0	0

23. 07-08 C3 Digital Electronics

The circuit translates to $\overline{AB+(B+C)}$ (0,0,0), (1,0,0)

$= \overline{AB}(\overline{B+C}) = (\overline{A}+\overline{B})(\overline{B}\,\overline{C}) = \overline{A}\,\overline{B}\,\overline{C} + \overline{B}\,\overline{C} = \overline{B}\,\overline{C}(\overline{A}+1) = \overline{B}\,\overline{C}$

$\overline{B}\,\overline{C} = 1 \rightarrow \overline{B} = 1$ & $\overline{C} = 1 \rightarrow B = 0$ & $C = 0$ & $A = *$

24. 07-08 C4 Digital Electronics

The circuit translates to: $(\overline{A}+(AB \oplus \overline{B+C}))CD$ (0,1,1,1), (0,0,1,1)

To be TRUE, $C = 1$ and $D = 1$ and $(\overline{A}+(AB \oplus \overline{B+C})) = 1$

If $C = 1$, then $(\overline{A}+(AB \oplus \overline{B+1})) = (\overline{A}+(AB \oplus \overline{1}))$

$= (\overline{A}+(AB \oplus 0))$

Now if $A = 1$, then $\overline{1}+(1B \oplus 0) = 0+(B \oplus 0)$

This is TRUE if $B = 1$ Hence (1, 1, 1, 1).

Now if $A = 0$, then $\overline{0}+(0B \oplus 0) = 1+(0 \oplus 0) = 1$

This is TRUE for $B = *$ Hence (0, *, 1, 1).

25. 08-09 C3 Digital Electronics

The circuit translates to $(\overline{A+\overline{A}B})((B+C)\overline{C})$ 0 or FALSE

$(\overline{A+\overline{A}B})((B+C)\overline{C}) = (\overline{A}\,\overline{\overline{A}B})(B\overline{C}+C\overline{C}) = \overline{A}(\overline{\overline{A}}+\overline{B})(B\overline{C})$

$= (A\overline{A}+\overline{A}\,\overline{B})(B\overline{C}) = \overline{A}\,B\,\overline{B}\,\overline{C} = 0$

26. 08-09 C3 Digital Electronics

The circuit translates to $\overline{\overline{A}(AB+\overline{C})}+(\overline{C+D})$ (0,1,0,1), (0,0,0,1)

$\overline{\overline{A}(AB+\overline{C})}+(\overline{C+D}) = \overline{A\overline{A}B+\overline{A}\,\overline{\overline{C}}}+\overline{C}\,\overline{D}$

$= \overline{\overline{A}\,\overline{\overline{C}}}+\overline{C}\,\overline{D} = A+C+\overline{C}\,\overline{D}$

If $A+C+\overline{C}\,\overline{D}=0$, then $A = 0 \wedge C = 0 \wedge \overline{C}\,\overline{D} = 0$.

$\overline{0}\overline{D} = 1\overline{D} = 0 \Rightarrow \overline{D} = 0 \Rightarrow D = 1$ Also $B = *$

27. 08-09 C4 Digital Electronics

The circuit translates to: $AB \oplus \overline{B\overline{C}}$ Choice B B, C, D

$AB \oplus \overline{B\overline{C}} = AB\overline{\overline{B\overline{C}}}+\overline{AB}\,\overline{B\overline{C}}$

$= ABB\overline{C}+(\overline{A}+\overline{B})(\overline{B}+C) = AB\overline{C}+\overline{A}\,\overline{B}+\overline{A}C+\overline{B}+\overline{B}C$

$= AB\overline{C}+\overline{B}+\overline{A}C$ Choice C

Choice D simplifies to Choice B using one of the Boolean identities.

28. 09-10 C3 Digital Electronics

The circuit translates to $\overline{((\overline{\overline{A}+B})+(BC))C}$ 5

$\overline{((\overline{\overline{A}+B})+(BC))C}=1 \Rightarrow (\overline{\overline{A}+B}+BC)C=0$

$(A\overline{B}+BC)C=0$

$A\overline{B}C+BC=0$

$if\ C=0\ then\ A=*\wedge B=*$

$if\ C=1\ then\ A\overline{B}+B=0$

$\Rightarrow \mathrm{B}=0\wedge \mathrm{A}=0$

That gives you 5 ordered triples: (1,1,0),(1,0,0),(0,0,0),(0,1,0),(0,0,1).

29. 09-10 C3 Digital Electronics

The circuit translates to $((\overline{AB})(\overline{B+C}))\square\ (C\oplus D)\ \square D$ 4

A	B	C	D	$\overline{AB}$	$\overline{B+C}$	$(\overline{AB})(\overline{B+C})$	$C\oplus D$	$\square$
0	0	0	1	1	1	1	1	1
0	1	0	1	1	0	0	1	1
1	0	0	1	1	1	1	1	1
1	1	0	1	0	0	0	1	1

Space did not allow the whole truth table. So here are the rows that made the expression TRUE. There are 4 of them.

30. 09-10 C4 Digital Electronics

The circuit translates to: $\overline{\overline{A+B}(\overline{B}C\oplus(C+D))}$ 1

To be FALSE, $\overline{A+B}(\overline{B}C\oplus(C+D))=1$

$\Rightarrow \quad \overline{A+B}=1\wedge \overline{B}C\oplus(C+D)=1$

$\overline{A+B}=1\Rightarrow A+B=0\Rightarrow A=0\wedge B=0$

So $(1C\oplus(C+D)=1$

Now if $C=1$, then $1+D=0$ which is impossible

Now if $C=0$, then $D=1$

$\therefore$ 1 ordered quadruple makes the expression FALSE

31. 10-11 C3 Digital Electronics

The circuit translates to $\overline{A}(A \oplus B) + \overline{\overline{BC} \oplus C}$. 4

There are 4 that make it TRUE.

A	B	C	$\overline{A}$	$A \oplus B$	$\overline{A}(A \oplus B)$	$\overline{BC}$	$\overline{\overline{BC} \oplus C}$	$\overline{A}(A \oplus B) + \overline{\overline{BC} \oplus C}$
0	0	0	1	0	0	1	0	0
0	0	1	1	0	0	1	1	1
0	1	0	1	1	1	1	0	1
0	1	1	1	1	1	0	0	1
1	0	0	0	1	0	1	0	0
1	0	1	0	1	0	1	1	1
1	1	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0

32. 10-11 C3 Digital Electronics

This circuit translates to $\overline{(AB + \overline{B}) + \overline{B + C}}$ $\overline{A}B$

$\overline{(AB + \overline{B}) + \overline{B + C}} = \overline{AB + \overline{B}\,\overline{B + C}} = (\overline{AB})\overline{\overline{B}}(B + C) = (\overline{A} + \overline{B})B(B + C)$

$= (\overline{A}B + \overline{B}B)(B + C) = \overline{A}B(B + C) = \overline{A}B + \overline{A}BC = \overline{A}B(1 + C) = \overline{A}B$

33. 10-11 C4 Digital Electronics

The circuit translates to: $\overline{(A + (A + B)C)(CD + D)}$ 5

$\overline{(A + (A + B)C)(CD + D)} = \overline{(AC + BC + A)(CD + D)}$

$= \overline{(A(C + 1) + BC)(D(C + 1)} = \overline{(A + BC)D}$

$= \overline{AD + BCD}$

The Truth Table for the above produces 5 FALSE results.

34. 11-12 C4 Digital Electronics

$\overline{A(A \oplus B)} + (B + C)C$ $= \overline{A} + \overline{A \oplus B} + BC + C$ (1,0,0)

$= \overline{A} + \overline{A} \oplus B + BC + C$

$= \overline{A} + \overline{A}\,\overline{B} + \overline{\overline{A}}B + BC + C$

$= \overline{A}(1 + \overline{B}) + AB + C(B + 1)$

$= \overline{A} + AB + C$

$\overline{A} + AB + C = 0 \Rightarrow \overline{A} = 0 \wedge AB = 0 \wedge C = 0 \Rightarrow$

$A = 1 \wedge B = 0 \wedge C = 0$

35. 11-12 C4 Digital Electronics

$((\overline{(A+B)(BC)})(\overline{C+D}) = (\overline{A+B}+\overline{BC})(\overline{C}\,\overline{D}) = (\overline{A}\,\overline{B}+\overline{B}+\overline{C})(\overline{C}\,\overline{D})$ $\overline{C}\,\overline{D}$

$= \overline{A}\,\overline{B}\overline{C}\,\overline{D}+\overline{B}\overline{C}\,\overline{D}+\overline{C}\,\overline{D} = \overline{C}\,\overline{D}$

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36. 12-13 C4 Digital Electronics

The circuit translates to: $\overline{(AB)(B+C)+\overline{\overline{C}}}$ (0,0,1), (0,1,1)

$\overline{(AB)(B+C)+\overline{\overline{C}}} = \overline{(AB)(B+C)}C = (\overline{AB}+\overline{B+C})C$ (1,0,1)

$= ((\overline{A}+\overline{B})+\overline{B}\,\overline{C})C = \overline{A}C+\overline{B}C+B\overline{C}C = (\overline{A}+\overline{B})C$

So $(\overline{A}+\overline{B})C=1 \Rightarrow \overline{AB}=1 \wedge C=1 \Rightarrow AB=0 \wedge C=1$

Therefore (0, 0, 1), (0, 1, 1), (1, 0, 1) make the circuit true.

37. 12-13 C4 Digital Electronics

Circuit translates $(A+\overline{(A+B)(BC)}) \oplus ((\overline{C+D})\overline{D})$ and simplifies to: 6

$(A+\overline{B}+\overline{C}) \oplus \overline{C}\,\overline{D}$. Of the 16 possible, 6 make if FALSE.

A	B	C	D	$\overline{B}$	$\overline{C}$	$\overline{D}$	$A+\overline{B}+\overline{C}$	$\overline{C}\,\overline{D}$	$\oplus$
0	0	0	0	1	1	1	1	1	0
0	0	0	1	1	1	0	1	0	1
0	0	1	0	1	0	1	1	0	1
0	0	1	1	1	0	0	1	0	1
0	1	0	0	0	1	1	1	1	0
0	1	0	1	0	1	0	1	0	1
0	1	1	0	0	0	1	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1	0
1	0	0	1	1	1	0	1	0	1
1	0	1	0	1	0	1	1	0	1
1	0	1	1	1	0	0	1	0	1
1	1	0	0	0	1	1	1	1	0
1	1	0	1	0	1	0	1	0	1
1	1	1	0	0	0	1	1	0	1
1	1	1	1	0	0	0	1	0	1

38. 13-14 C4 Digital Electronics

The circuit translates to: $(\overline{AB})(\overline{B+C}) \oplus C$ (*.*.1), (*,0,0)

$(\overline{AB})(\overline{B+C}) \oplus C = (\overline{A}+\overline{B})(\overline{B}\,\overline{C}) \oplus C = (\overline{A}\,\overline{B}\overline{C}+\overline{B}\,\overline{B}\,\overline{C}) \oplus C = \overline{B}\,\overline{C} \oplus C$ (1,1,1),(1,0,1),

So $\overline{B}\,\overline{C} \oplus C = 1$. $C=1 \rightarrow \overline{B}\,\overline{C}=0 \rightarrow B=*, A=*, C=1$ So (*, *, 1) (0,1,1),(0,0,1),

$C=0 \rightarrow \overline{B}\,\overline{C}=1 \rightarrow B=0, A=*, C=0$ So (*, 0, 0) (1,0,0),(0,0,0)

39. 13-14 C4 Digital Electronics

Circuit translates $\overline{(\overline{A}+AB)+((\overline{B+C})(\overline{CD}))D}$ $A\overline{B}\overline{C}D$

$= (\overline{\overline{A}+AB})(\overline{\overline{(\overline{B+C})(\overline{CD})}})D = (\overline{\overline{A}}\,\overline{AB})((\overline{B+C})(\overline{CD})D$

$= A(\overline{\overline{A}+\overline{B}})((\overline{B+C})+\overline{CD})D = (AD\overline{A}+AD\overline{B})(\overline{B}\overline{C}+\overline{C}+\overline{D})$

$= AD\overline{B}\overline{B}\overline{C}+AD\overline{B}\overline{C}+AD\overline{B}\overline{D} = A\overline{B}\overline{C}D$

40. 14-15 C4 Digital Electronics

$\overline{(\overline{AB})(\overline{BC})}+(\overline{BC}B\oplus BC) = (\overline{\overline{AB}}+\overline{\overline{BC}})+(((\overline{B}+\overline{C})B)\oplus BC)$ 8

$= ((\overline{\overline{A}}+\overline{B})+BC)+((\overline{B}B+\overline{C}B)\oplus BC) = (A+\overline{B}+BC)+(\overline{B}B\oplus BB)$

$= (A+\overline{B}+BC)+(0\oplus B) = (A+\overline{B}+BC)+(0\overline{B}+\overline{0}B)$

$= (A+B+BC)+(0+1)$

$= (A+\overline{B}+BC)+1=1$

Therefore all triples will make it TRUE.

41. 14-15 C4 Digital Electronics

$\overline{(\overline{A}\oplus\overline{\overline{A}B})+(\overline{B\overline{C}+C})} = (\overline{\overline{A}\oplus\overline{\overline{A}B}})(\overline{\overline{B\overline{C}+C}})$ (0,0,1)

$= (\overline{A}\,\overline{\overline{AB}}+\overline{\overline{A}}\,\overline{A}B)(B\overline{C}+C) = (\overline{A}(\overline{\overline{A}}+\overline{B}))(B\overline{C}+C) = (\overline{A}A+\overline{A}\overline{B})(B\overline{C}+C)$

$= \overline{A}\overline{B}B\overline{C}+\overline{A}\overline{B}C = \overline{A}\overline{B}C$

Therefore (0,0,1) makes it TRUE.

42. 15-16 C4 Digital Electronics

The circuit translates to: $\overline{\overline{A}\,\overline{\overline{AB}}+(B+\overline{C})C}$ $A\overline{B}C$

$\overline{\overline{A}\,\overline{\overline{AB}}+(B+\overline{C})C} = \overline{\overline{A}(\overline{\overline{A}}+\overline{B})+(B+\overline{C})C} = \overline{\overline{A}A+\overline{A}\overline{B}+B+\overline{C}C} =$

$\overline{\overline{A}\overline{B}+B+\overline{C}C} = \overline{\overline{A}}\,\overline{\overline{B}}\,\overline{B}CC = (\overline{\overline{A}}+\overline{\overline{B}})\overline{B}C = A\overline{B}C+B\overline{B}C = A\overline{B}C$

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The circuit translates to: $\overline{\overline{(\overline{A}+BC)+(\overline{CD}+D)}}$ 0

$\overline{\overline{(\overline{A}+BC)+(\overline{CD}+D)}} = \overline{\overline{(\overline{A}+BC)}}\,\overline{(\overline{CD}+D)} = (\overline{A}+BC)(\overline{\overline{CD}}\,\overline{D})$

$= (\overline{A}+BC)CD\overline{D} = 0$